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Error Detection of Data Conversion in Flash ADC using Code Width Based Technique

Senthil Sivakumar M^{a*}, Gurumekala T^b, Sruthi Pulya^a^a*Vignan's Foundation of Science, Technology and Research, Guntur, Andhra Pradesh, India*^b*Madras Institute of Technology, Anna University, Chennai, India*

Abstract

The high integration density of the complex electronic system requires the multi-functional testing facility to ensure the accuracy of the circuit function. In addition, the wide population of submicron technologies results in the persistent requirements of high precision analog and mixed-signal (AMS) circuits. In the complex, high-density circuits, observing the correctness of output are limited due to the limitations of input and output pins which develop the AMS circuit test as very difficult and expensive. A digital built-in self-test (BIST) scheme has been presented in this paper for testing an analog to digital converter (ADC) in the time domain. The testing scheme consists of the linear analog ramp generator, ADC as a circuit under test, output response analyzer and a BIST controller. Bootstrap linear ramp generator is used to generate the linear analog ramp signal which is applied to ADC to develop the digital word sequence for testing. Among various types of ADCs, here a Flash ADC has used since it is fastest and accurate in digital conversion. A Flash ADC has been designed with the seven-bit resolution and tested using the proposed digital BIST scheme. The ADC comprises a sample-hold circuit, $2^N - 1$ comparator, XOR gates, and encoder block. The output response analyzer verifies the digital output sequence to validate the static test parameters of ADC called monotonicity, missing codes, DNL, and INL error. The BIST controller generates the control signals to control the test sequence. The complete test sequence of ADC BIST has simulated in Tanner EDA with TSMC 0.18 μ m technology. ORA is used to analyze the presence of monotonicity, missing codes, DNL and INL error in the ADC output.

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* Corresponding Author. Tel.: +91-9380984544

E-mail address: msenthilsivakumar@gmail.com

1. INTRODUCTION

The Analog to Digital Converter (ADC) is an indispensable and very important mixed-signal block in various systems and applications [1]. The use of digital signal processing (DSP) techniques [2] in the modern electronic systems for high performance and processing accuracy over analog signal processing demands the ADC to strap with the real-time signal. An analog signal has been processed, stored and transmitted to the DSP processors in digital form for high-speed data processing [3]. The ADCs are used in the microcontrollers typically with the resolutions of 8-bits, 10-bits, 12-bits, and 16-bits. In the recent development of the electronic systems, the analog signals are converted into digital form for most advanced signal processing tasks that have been used in the signal processing applications. The ADC circuit develops a liaison from the real-time analog input signals or a system to the digital output signals or the digital processors [4]. The testing of the high-speed data converters ensures that the testing technique supports other types of ADCs also without any performance degradation in output [5]-[6].

The parallel processing technique used in the flash type ADC utilizes high power consumption with the high-resolution data conversion due to the use of more number of comparators and voltage dividers. Several data conversion techniques have been proposed for high-resolution flash type ADC for low power consumption [7]. A folding and averaging conversion technique with two-stage flash ADC has been designed and presented in [8] with the intention of improving the performance in power utilization. The standard two-stage analog comparators consume less power in the implementation of flash type ADC with less sensitization in noise [9]-[10]. Comparator is the essential circuit in the flash ADC for low power application. The use of a low power comparator circuit in the flash type ADC architecture [11]. Threshold inverter quantization, two-stage folded comparators, dynamic comparator, clock-based double-tail comparators are the popular comparator structures used in the flash ADC. The dynamic comparators are in the flash ADC to provide a strong feedback path through the regenerative latch [12].

Testing of the high-speed ADC is the most important requirement in the AMS type electronic applications. The 6-bit flash type ADC has been designed and tested in 0.18 μ m CMOS technology. The design of ADC is observed for the analog input voltage range of 360 mV to 560 mV. A high-speed medium resolution flash type ADC has been designed and tested in [13] for ultra-wideband applications. A 4-bit flash type ADC design has been proposed in [14] with testing of the data transitions in the data conversion. The primary results are verified in 130nm CMOS technology with online analog ramp generators for built-in self-test applications. The fully differential type ramp generator has been used and integrated with ADC for digital test pattern generation. The conversion results are tested for the static characteristics of the ADC. The on-chip ramp generator is targeted for efficient implementation of BIST for ADC testing with static characteristic responses. This paper presents the testing of the flash type ADC with the resolution of 7-bit resolution using digital BIST techniques.

2. TEST CHARACTERISTICS OF ADC

ADC is a mixed-signal circuit that defines the analog reference voltages in digital form through the sampling and holding process of an analog input signal [15]. The circuit of ADC accepts the continuous-time analog input signal and compared with the analog reference voltages to deliver the digital response as shown in Fig. 1.

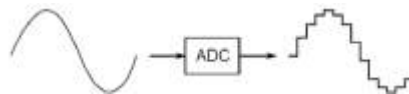


Fig. 1. Block diagram of the dynamic comparator based flash ADC.

The continuous-time analog signal is sampled into discrete N-bit digital word in the ADC as expressed in equation (1).

$$V_i = V_{FS} \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}} + \varepsilon \quad (1)$$

The voltage difference of two adjacent analog reference voltage points is referred as the quantum voltage level (V_q) and it is expressed as in equation (2).

$$V_q = \frac{V_{FS}}{2^N} = 1.LSB \quad (2)$$

The change reference switching position in the analog signal results the change in the position of the digital output response [16] and they get change with respect to the quantization process inherent to the conversion [17] as shown in Fig. 2. The change of reference voltage in the ADC is expressed in equation (3) as follows

$$\Delta V = \frac{V_r}{2^N} \quad (3)$$

Where V_r is the reference voltage point, N is the resolution of the ADC in, 2^N is number of bits, ΔV is the resolution.

In an ADC testing, the linear analog ramp signal is applied as an input. The developed digital output equivalent to the analog amplitude is analyzed for measuring the static parameters as shown in Fig. 2 (a) & (b). DNL is the difference in the width of actual code from ideal code as shown in equation (4),

$$DNL(K) = \frac{V_{K+1} - V_K}{V_{LSB}} - 1.LSB \quad (4)$$

Where, V_{LSB} is the quantization step, V_K , V_{K+1} are the reference voltage at K^{th} and $(K+1)^{th}$ positions respectively. Integral non-linearity (INL) error is a deviation between the actual and the ideal code as expressed in equation (5).

$$INL(K) = \frac{\sum_{i=0}^{K-1} V_i - K.LSB}{1.LSB}, \quad \text{where, } K \rightarrow 0, 1, 2, \dots, (2^N - 1) \quad (5)$$

The offset and gain error of the ADC are obtained as shown in equation (6) & (7) respectively.

$$Offset_error = \frac{V_{AMIN} - V_{IMIN}}{V_{LSB}} \quad (6)$$

$$Gain_error = \left(\frac{V_{AMAX} - V_{AMIN}}{V_{IMAX} - V_{IMIN}} - 1 \right) \times 100 \quad (7)$$

Where V_{AMAX}/V_{IMAX} and V_{AMIN}/V_{IMIN} are the last and first actual/ideal reference voltage respectively, In an ADC, if the output code always increases and decreases with respect to the ADC input, this is called monotonicity. The presence of missing code is noted when the DNL value is greater than 1LSB.

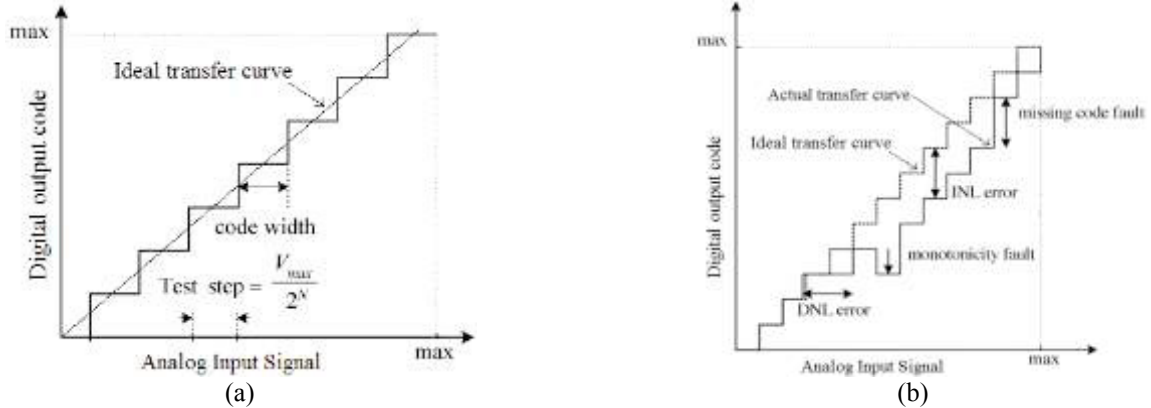


Fig. 2. Transfer Characteristics of an ADC (a) Code Width Measurement; (b) Static Characteristic Parameters.

3. PROPOSED ADC TESTING

The general structure of the proposed ADC BIST is shown in Fig.3.

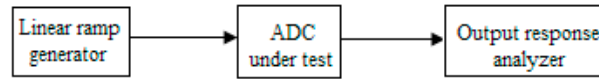


Fig. 3. The test structure of the proposed ADC.

3.1. Bootstrap Linear Ramp Generator

Linear input signals are generally used for testing an ADC. Linear analog ramp generators are most widely used in the testing of the ADC. The linearity of the ramp signal dominates the accuracy testing. The fault model of the ADC test methodology can be tested in the time domain using the ramp signal as easier and precision due to its smaller area overhead utilization and shorter testing time. The constant increment of the ramp slope in by charging the capacitor using low gain current source provides linear analog ramp signal more suitable for testing based applications. The bootstrap generator has been used to generate the highly linear positive ramp slope. The schematic of the ramp generator is shown in Fig. 4. This circuit generates the linear ramp signal by controlling the transistors Mp3 and Mn2 using the input pulse that enables the charging and discharging of the capacitor. The input pulse is applied through an inverter which provides full swing through the transistors Mp3 and Mn2. The charge capacitors C1 and C2 are charged and discharged through the feedback output voltage and controlled through the inverted pulse. The constant current source developed by the feedback configuration provides the linear ramp.

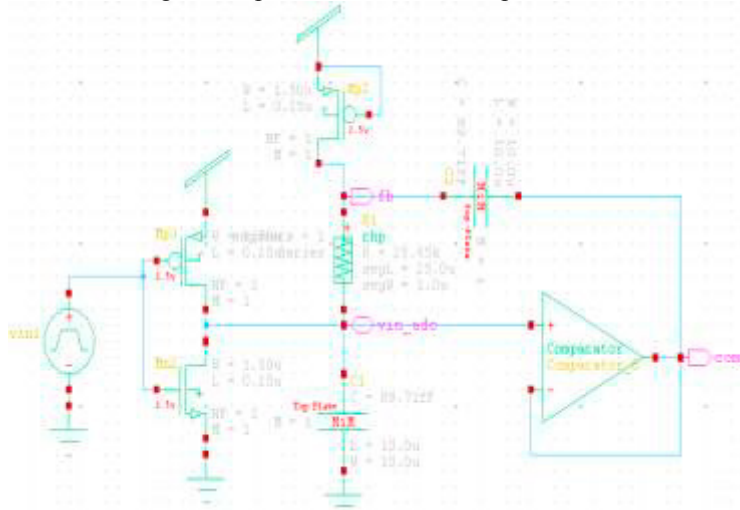


Fig. 4. A bootstrap linear ramp generator.

3.2. Flash ADC

The structure of the flash type ADC provides fast data conversion. An ADC structure shown in Fig.5 has been implemented and used for testing. The circuit is implemented using a sample/hold (S/H) circuit, preamplifier, latch track comparator, and an encoder [18]. The parallel comparators used in the ADC responds instantaneously to the present input, and provide the digital output equivalent to the analog magnitude. The comparator time constant becomes high which limits the speed, particularly in track mode. Mathematically, it is expressed in equation (8),

$$\tau_{latch} = \frac{C_{tot}}{g_m} \ln \left(\frac{\Delta V}{\Delta V_o} \right) \quad (8)$$

Where g_m is the trans-conductance of the transistor, C_{tot} is the gate to source capacitance of the transistor, ΔV is the voltage difference between output voltages of the inverter, ΔV_o is the initial voltage difference at latching.



Fig. 5. Structure of Flash ADC.

3.3. Output Response Analyzer

The Output Response Analyzer (ORA) is made up of delay elements, subtractors, and stepsize counters are shown in Fig 6. ORA is used for comparing the output and it is used to avoid the errors in ADC before the fabrication process. For identifying the consecutive changes of output delay element used with one clock delay. Adder/Subtractor is used to identify the difference of the actual output with the delayed output. DFF is used for delaying the output from the digital encoder for further process. Clk is used to store the signal for the desired time which will be given to the next block. The reset is used to clear the information. The block diagram of the ADC testing is shown in Fig 6. The reset or clear should always be zero for the continuous-time of the signal. Recent data from both the ADC and delay element is given to the subtractor which gives the difference output. The logic OR of output difference bits measure the missing code. The sign bit represents a monotonic fault. The difference from each stage is taken out simultaneously for input to the next block is shown in Fig 6. Borrow bit the monotonic fault. Another register and subtractor block is used to find out the DNL and INL error. Active low in all the bits of difference is no transition which will make the register output as zero. Thus the register output is again subtracted by the code width counter output for indicating the DNL error. The serial addition of the DNL error denotes the INL error. The same procedure is repeated again with the generated reset from the first testing block.

4. Simulations and Result Analysis

The test implementation structure of the proposed ADC test has been simulated and tested in TSMC 180nm CMOS technology. The bootstrap ramp generator develops the linear ramp signal by the use of a constant current source. Generated ramp signal applied to the ADC to derive the logic components. The comparators utilized in the flash type ADC develops the digital form of output by sampling the ramp signal without any delay or distortion. Fig.7 shows the linear ramp signal generated by the current source designed in the bootstrap ramp generator. Bootstrap ramp generator is generating the ramp signal by linear charging of the capacitor. The bootstrap ramp circuit is employed in the BIST along with ADC for testing the data conversion. The accuracy of the ramp signal improves the accuracy of data conversion in discrete test pattern generation.

A 7-bit ADC has been driven by the analog ramp input to develop digital word as shown in Fig. 8. The digital testing blocks of ORA has been implemented in HDL and simulated for error detection. A delay element used in ORA produces one clock delay to a digital word. The actual digital core is compared with the one clock shifted digital code in the subtractor to find the difference output as shown in Fig. 9. Using the difference output missing code can be calculated and mc_0 denotes the missing code fault along with the $diff0_LSB$. The fault of the ADC observed as the difference from subtractor is deviate from the value $0...0001$. The logic OR of difference output s_1 to s_{n-1} detects the missing code fault. Missing code fault detection is shown in Fig.8. At any time, the difference between the successive output codes would diverge by 1.LSB then borrow should be one. Monotonicity of an ADC can be calculated by using $borrow_mono_1$ from the Fig.8. It is clear that the monotonicity fails in the time period (0.82micro) and (1.15micro). If monotonicity fails, the code transition also fails. The glitches or distortion in the output signal is due to the gate delay. The final borrow indicates the monotonicity of the ADC. Mc_0 indicates missing code should be zero always and $borrow_mono_1$ shows that the transition is monotonically converted. The step size counter counts transition length terms of clock cycles. The counter resets into the initial position when the transition point is identified. The code width is measured from the transition length repeatedly for each digital code observed from ADC. A reset calculated from the previous block is used for INL and DNL calculations. It is given to the second register, the output of this register is shown in Fig. 9 and it is compared with the actual ADC output is shown in Fig. 9. DNL can be measured from the percentile of the maximum deviation measured by comparing it with the ideal step size. In Fig. 9,

the DNL error or code width fault is shown. Due to the monotonicity of the ADC output fails in (0.82micro) and (1.15micro), the code width variation occurred at the same time period. It can be said to be a DNL error. Measured DNL errors can be used to calculate the INL error. The serial adder is used for calculating the INL error from the DNL error. In this Fig. 9, the addition of those two code width fault defines the INL error. It provides information about the offset error and gains error. Finally, if there is a presence of missing code, code width fault, INL and failure in monotonicity in a circuit is said to be a faulty circuit. The comparison table of BIST techniques is shown in Table.1.

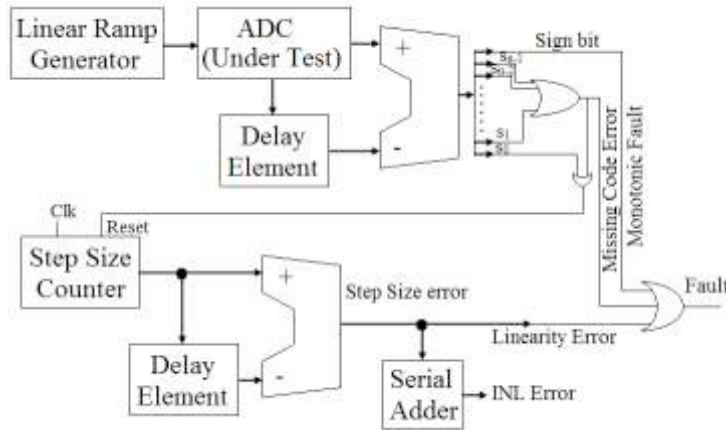


Fig. 6. Block diagram for the BIST for finding the missing code, monotonicity, DNL, and INL error.

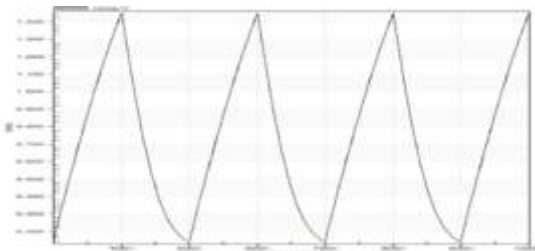


Fig. 7. The output response of Bootstrap linear ramp generator.

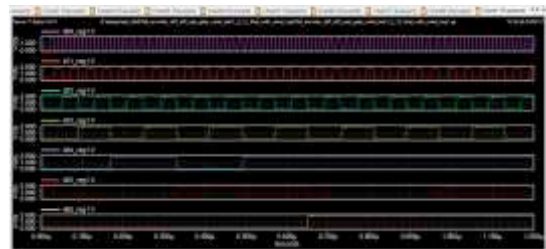


Fig. 8. ADC output.

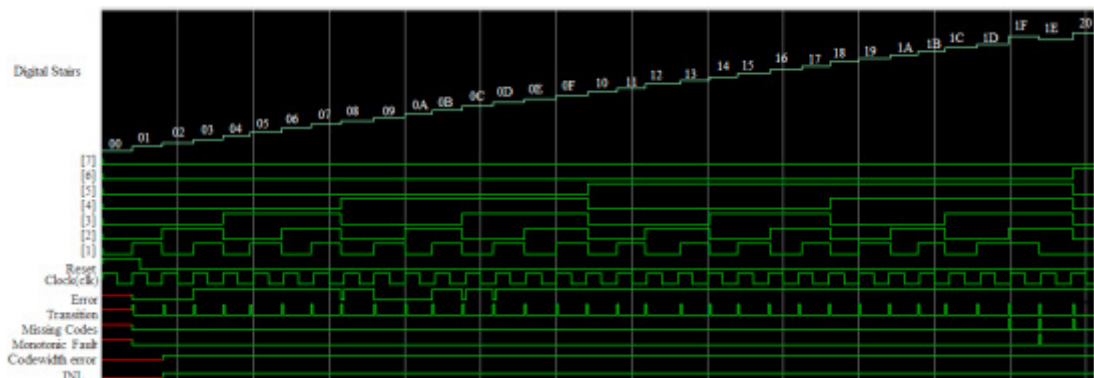


Fig. 9. Identification of Code width fault, Missing code, monotonic fault, INL

Table. 1 Synthesis result of 7-bit ADC BIST

Technique	Adder	Subtractor	Counter	Flip-Flops	Multiplexers	Logic gates	Decoder
Ref.[9]	2	4	2	24	-	OR –2	-
Ref.[4]	2	4	2	20	-	OR –2	-
Ref.[7]	3	4	3	28	8:1– 1	-	-
Ref.[18]	2	1	2	8	2:1 – 16 8:1 –1	Ex-OR-2 Ex-NOR-1	4:16
This Work	1	2	1	12	-	OR –2	-

5. Conclusion

In this project, a design of 7-bit Flash ADC BIST has been proposed and tested using Tanner EDA. A bootstrap ramp generator employed in this testing scheme generated a linear ramp signal which develops a 7-bit sequence of digital output from ADC. The sequential code analysis has been performed by the output response analyzer on the digital output sequence. The static parameters of ADC like linearity's, missing codes, and monotonicity are verified by the proposed digital test architecture. The test results are obtained and verified through simulation. The digital-based test system implemented in this work is very accurate when the digital test patterns are generated accurately by the ramp generator. The linear ramp based ADC test system improves the accuracy of testing and greatly reduces the testing time and cost. On-chip testing in ADC increases the feasibility testing characteristics with respect to the operating performance. Effective on-chip digital BIST technique is used for testing reduces the area consumption. The future work involves testing the ADC characteristics, the counter will enhance the testing performance. The counter is used to detect the quantization error while converting the analog signal into digital. It generates the original signature for comparing with the ADC.

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